

MAT 2306- Operations Research.

Grading System

Attendance 10% -

Open Book Test 20%

Home work 10%

End Exam 60%

Operations Research (OR)

The term OR has its origin during the 2nd world war. This has an interesting history refer to the text book.

Introduction to linear programming.

This is constrained optimization technique, which optimizes criterion within same constraints. In linear programming, the objective function (Profit, loss or return on investment) are constant and are linear. There are different methods available to solve linear programming.

Definition 1

A function $f(x_1, x_2, \dots, x_n)$ of the variables $x_1, x_2, \dots, x_n$ is a linear function if

$$f(x_1, x_2, \dots, x_n) = C_1x_1 + C_2x_2 + C_3x_3 + \dots + C_nx_n$$

Where $C_1, C_2, \dots, C_n$ are constants.

Example

$f(x_1, x_2) = 2x_1 + x_2$ is linear function while $f(x_1, x_2) = x_1^2 x_2$ is not linear.

Defⁿ

for any linear function $f(x_1, x_2, \dots, x_n)$ and any real number b ,

The inequalities $f(x_1, \dots, x_n) \leq b$ and $f(x_1, x_2, \dots, x_n) \geq b$ are linear inequalities.

Example:

$2x_1 + 3x_2 \leq 3$ is linear inequality while
 $2x_1, x_2 + \sqrt{x_2} \geq 3$ is not linear inequality.

Def.ⁿ

A linear programming (LP) problem is an optimization problem for which are do the following (General structure of an LP model)

- 1). We attempt to maximize (Profit) or minimize (cost) a linear function (called objective function) of the decision variables.
- 2). The values of the decision variables must satisfy a set of constants and each constraint must be linear equation or linear inequalities.
- 3). A sign restriction is associated with each variable for any variable either $x_i \geq 0$ or x_i is unrestricted in sign $\max(\text{or min}) Z = f(x_1, x_2, \dots, x_n)$
Subject to constraints sign restriction.

Formulation of the linear programming Model.
Three components are,

- 1). The decision variable.
- 2). The environment (uncontrollable) parameters
- 3). The result (dependent) variable.

General Mathematical Model of linear programming Problem.

The general linear programming problem (or model) with n decision variables and m constraints can be stated in the following form:

Optimize (Max or min) $Z = C_1x_1 + C_2x_2 + \dots + C_nx_n$
Subject to the linear constraints.

$$\begin{aligned}
 & a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq, =, \geq) b_1 \\
 & a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq, =, \geq) b_2 \\
 & \vdots \\
 & a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n (\leq, =, \geq) b_m \text{ and} \\
 & x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0.
 \end{aligned}$$

The above formulation can also be expressed in a compact form as follows.

$$\text{Optimize (Max or min)} \quad z = \sum_{i=1}^n c_i x_i \quad (\text{Objective function})$$

$$\text{Subject to the linear constraints} \\
 \sum_{j=1}^n a_{ij} x_j (\leq, =, \geq) b_i \quad ; \quad i = 1, 2, \dots, m$$

Where c_i 's are the coefficients representing the per unit profit (or cost) of decision variable x_j of the value of objective function. The a_{ij} 's are referred as technological coefficients. These represent the amount of resources say i consumed per unit of variable x_j . These coefficients can be positive, negative or zero. The b_i represents the total availability of the resource. The term resource is used in a very general sense to include any numerical value associated with the right hand side of a constraint.

It is assumed that $b_i \geq 0$ for all i . However if any $b_i < 0$. Then both sides of constraint i is multiple by -1 to make $b_i > 0$ and reverse the inequality of the constraints.

Example 1.

A firm produces the 3 products these products are processed on the different machine the time required to manufacture on unit of 3 product and daily capacity of the 3 machine are given in the table below.

Machine	Time per unit (minutes)			Machine capacity (minutes/day)
	Product 1	Product 2	Product 3	
M ₁	2	3	2	440
M ₂	4	—	3	470
M ₃	2	5	—	430

It is required to determine the daily number of units to be manufactured for each product. The profit per unit for product 1, 2 and 3 is Rs. 4, Rs. 3 and Rs. 6 respectively. It is assumed that all the amount produced are consumed in the market formulate the mathematical (LP) model that will maximize the daily profit.

formulation of LP

steps.

- 1). Identify the decision variables.
- 2). Write objective function.
- 3). formulate constraints.

formulation of linear programming model.

steps.

- 1). From the study of the situation find the key decision to be made in the given situation key decision is to decide the extent of product 1, 2 and 3 as the extents are permitted to vary.
assume symbol for variable quantity notice in

- 2). Let the extent of product 1, 2 and 3 manufactured daily be x_1, x_2, x_3 units respectively.
- 3). Express the feasible alternatives mathematically. Feasible alternative are those which are physically and economically in the given situation. Feasible alternatives are set of values of x_1, x_2 and x_3 units respectively.

Since negative production has no meaning and is not feasible.

- 4). Maintain the objective function quantitatively and in the present situation objective to maximize the profit that,

$$\max Z = 4x_1 + 3x_2 + 6x_3$$

- 5). What are the constraints?

$$2x_1 + 3x_2 + 2x_3 \leq 440$$

$$4x_1 + 0x_2 + 3x_3 \leq 470$$

$$2x_1 + 5x_2 + 0x_3 \leq 430$$

Example 2

A factory manufactures two products A and B. To manufacture one unit of A, 1.5 machine hours and 2.5 labour hours are required. To manufacture product B, 2.5 machine hours and 1.5 labour hours are required. In a month, 300 machine hours and 240 labour hours are available. Profit per unit for A is Rs. 50 and B is Rs. 40. Formulate LP.

Products	Resource unit	
	Machine	Labour.
A	1.5	2.5
B	2.5	1.5
Availability	300 hrs	240 hrs

x_1 = Number of units of A manufactured per month

x_2 = " " " " B " "

$$\max Z = 50x_1 + 40x_2$$

$$1.5x_1 + 2.5x_2 \leq 300$$

$$2.5x_1 + 1.5x_2 \leq 240 \quad \text{and} \quad x_1 \geq 0, x_2 \geq 0.$$

Example 03.

A company owns two flour mills A and B, which have different production capacity for high, medium and low quality flour. The company has entered a contract to supply flour to a firm every month with at least 8, 12 and 24 quintals of high, medium and low quality respectively. It costs the company Rs. 2000 and Rs. 1500 per day to run mill A and B respectively. On a day mill A produces 6, 2 and 4 quintals of high, medium and low quality flour. Mill B produces 2, 4 and 12 quintals of high medium and low quality flour respectively. How many days per month should each mill be operated in order to meet the contract order most economically.

Let us define x_1 and x_2 are the mills A and B.

$$\text{Min } Z = 2000x_1 + 1500x_2$$

$$6x_1 + 2x_2 \geq 8$$

$$2x_1 + 4x_2 \geq 12$$

$$4x_1 + 12x_2 \geq 24$$

$$x_1, x_2 \geq 0$$

Example 04.

Farmer Johns must determine how many acres of corn and wheat to plant this year. An acre of wheat yields 25 bushels of wheat and requires 4 hours of labour per week. An acre of corn yields 10 bushels of corn and requires 4 hours labour per week. All wheat can be sold at 4 a bushel and all corn can be sold at 3 a bushel. Seven acres of land and 40 hours per week of labour are available. Government regulations require that at least 30 bushels of corn be produced during the current year. Formulate and solve the LP whose solution will tell farmer Jones how to maximize the total revenue from wheat and corn.

x_1 = number of acres of wheat planted.
 x_2 = " " " " corn "

$$\max Z = 4x_1 + 3x_2$$

$$\frac{1}{25}x_1 + \frac{1}{10}x_2 \leq 7$$

$$x_2 \geq 30.$$

Example 5

Company produces three products A, B, C for manufacturing three raw materials P, Q, R are used profit per unit,
 A - Rs. 5, B - Rs. 3, C - Rs. 4 resource requirement & limit.

Raw material \ Product	P (x_1)	Q (x_2)	R	maximum raw material availability.
A (x_1)	-	20	50	P - 80 units
B (x_2)	20	30	-	Q - 100 "
C (x_3)	30	20	40	R - 150 "

x_1 = Product of A

x_2 = Product of B

x_3 = Product of C.

$$\max = 5x_1 + 3x_2 + 4x_3$$

$$P \Rightarrow x_1 + 20x_2 + 30x_3 \leq 80$$

$$Q \Rightarrow 20x_1 + 30x_2 + 20x_3 \leq 100$$

$$R \Rightarrow 50x_1 + x_2 + 40x_3 \leq 150$$

$$x_1, x_2, x_3 \geq 0.$$

Solution to linear programming problem (LPP)

Feasible solⁿ

A set of real values $x = (x_1, x_2, \dots, x_n)$ which satisfies the constraint.

$[Ax (\leq, =, \geq) b]$ and satisfies the non-negative restriction $x \geq 0$.

Feasible Region.

The collection of all the feasible solution is called as the feasible region.

* When a LP is solved, one of the following cases will occur.

case I - The LP has unique solution when has only one optimal solution.

case II - The LP has more than one optimal solutions. This is the case of Alternative (Multiple) optimal solⁿ.

case III - The LP is infeasible i.e. has no feasible solⁿ that means that the version contains no points.

case IV - The LP is unbounded this means that there are points in the feasible region with arbitrarily large z -values.

There are many methods to solve LPP.

Graphical Method.

example.

1). Solve the following LPP graphically.

$$\max Z = 4x + 5y$$

$$\text{Subject to } x + y \leq 20$$

$$3x + 4y \leq 72$$

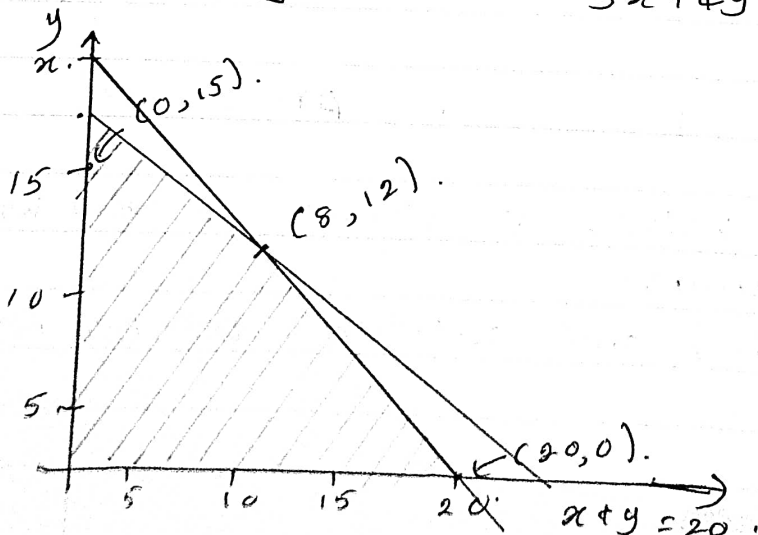
$$x, y \geq 0$$

$$x + y = 20$$

$$3x + 4y = 72$$

$$\Leftarrow x + y \leq 20$$

$$3x + 4y \leq 72$$



$$(0,0) \Rightarrow z = 0$$

$$(0,15) \Rightarrow z = 75$$

$$(8,12) \Rightarrow z = 72$$

$$(20,0) \Rightarrow z = 80$$

corner points

(20, 0)

(0, 18)

(8, 12)

Value of z

80

90

92

← The minimum value of the objective function.

finally, optimal solⁿ $x=8$ and $y=12$ which is unique

2). Solve the following LPP by graphical method,
Max $z = 100x_1 + 40x_2$

Subject to $5x_1 + 2x_2 \leq 1000$

$3x_1 + 2x_2 \leq 900$

$x_1 + 2x_2 \leq 500$

$x_1, x_2 \geq 0$

$x_1=0, x_2=500$ (0, 500)

$x_2=0, x_1=200$ (200, 0)

$x_1=0, x_2=450$ (0, 450)

$x_2=0, x_1=300$ (300, 0)

$x_1=0, x_2=250$ (0, 250)

$x_2=0, x_1=500$ (500, 0)

max z { (200, 0) = 20000
(0, 250) = 10000
(125, 187.5) = 20000 }

maximum value of objective funⁿ z.

finally there are multiple optimization solⁿ of LPP
(0, 250) (125, 187.5)

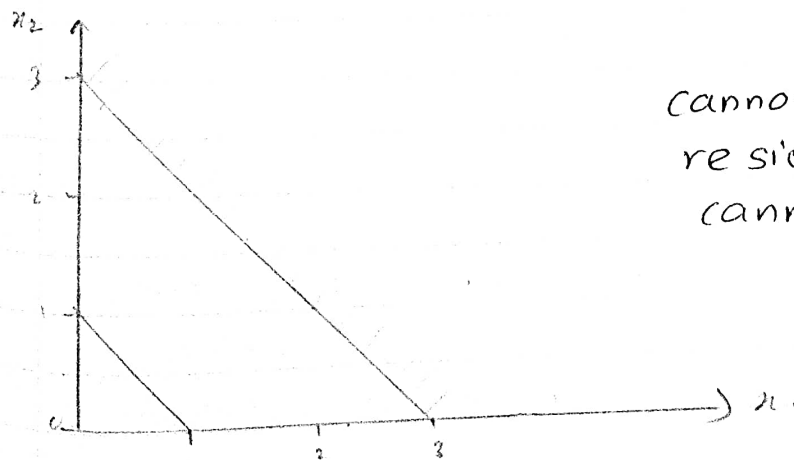
3). Max $z = x_1 + x_2$

$x_1=0, x_2=1$ (0, 1)

$x_2=0, x_1=1$ (1, 0)

Subject to $x_1 + x_2 \leq 1$

$x_1=0, x_2=3$ (0, 3)
 $x_2=0, x_1=3$ (3, 0)
 $x_1 + x_2 \geq 3, x_1, x_2 \geq 0$



cannot find a feasible region. so the problem, cannot be solve.

A.w.

Use graphical method to solve following LP
 Max $z = 3x_1 + 2x_2$ s.t. $5x_1 + x_2 \geq 10$

$$x_1 + x_2 \geq 6$$

$$x_1 + 4x_2 \geq 12 \quad x_1, x_2 \geq 0$$

$$x_1 + 4x_2 \geq 12$$

$$x_1 = 0 \quad x_2 = 3 \quad (0, 3)$$

$$x_2 = 0 \quad x_1 = 12 \quad (12, 0)$$

$$5x_1 + x_2 \geq 10$$

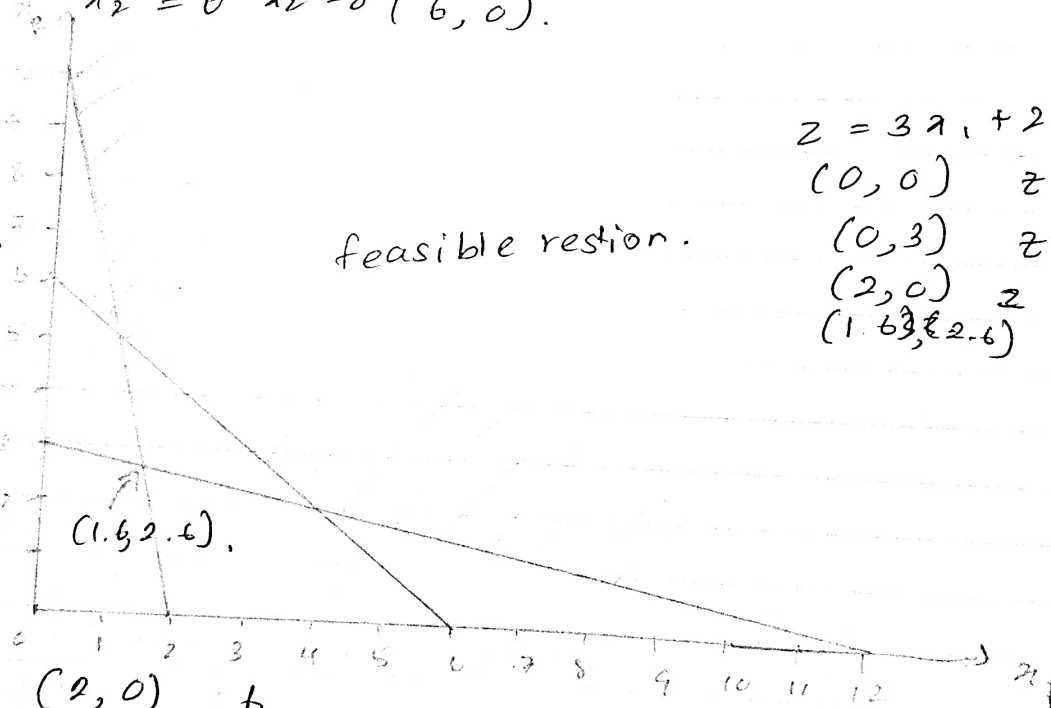
$$x_1 = 0 \quad x_2 = 10 \quad (0, 10)$$

$$x_1 = 2 \quad x_2 = 0 \quad (2, 0)$$

$$x_1 + x_2 \geq 6$$

$$x_1 = 0 \quad x_2 = 6 \quad (0, 6)$$

$$x_2 = 6 \quad x_1 = 0 \quad (6, 0)$$



feasible region.

$$z = 3x_1 + 2x_2$$

$$(0, 0) \quad z = 0$$

$$(0, 3) \quad z = 6$$

$$(2, 0) \quad z = 6$$

$$(1.6, 2.6) \quad z = 10$$

$$(2, 0) \quad 6$$

$$(0, 3) \quad 6$$

$$(1.6, 2.6) \quad 10 \leftarrow \text{maximum value of objective function}$$

finally optimal solⁿ $x = 1.6$ and 2.6 which is unique.

The feasible region is unbounded. Thus the maximum value of z occurs at infinity.

$\therefore$ Problem has unbounded solution.

Table 2. The Simplex Method.
 Examine $z_j - c_j$'s. If all are ≥ 0 , stop the operation optimum solution is reached. Then consider that x_j for which $z_j - c_j$ is the least. If $z_j - c_j = 0$ all basic and non basic variable, then multiple solution exist from the LPP. consider the ratio $\left\{ \frac{x_B}{y_{ij}} : y_{ij} > 0 \right\}$. If all the $y_{ij} \leq 0$, stop the iteration. LPP has unbounded solution.

$$\text{Key column} = \min (z_j - c_j)$$

$$\text{Key row} = \min \left(\frac{x_B}{y_{ij}} \right)$$

$$Z_j = C_B \cdot x_1 + C_B \cdot x_2 + C_B \cdot x_3$$

$$= C_B \cdot x_1 + C_B \cdot x_2 + C_B \cdot x_3$$

Example 2.

Solve the following LPP by simplex method.

$$\text{Max } Z = 2x_1 + 4x_2$$

$$\text{Subject to, } 2x_1 + 3x_2 \leq 48$$

$$x_1 + 3x_2 \leq 42$$

$$x_1 + x_2 \leq 21 \quad \text{and } x_1, x_2 \geq 0.$$

Solution.

Let x_3, x_4 and x_5 are slack variables with cost coefficient '0'.

$$Z = 2x_1 + 4x_2 + 0x_3 + 0x_4 + 0x_5$$

$$2x_1 + 3x_2 + x_3 + 0x_4 + 0x_5 = 48$$

$$x_1 + 3x_2 + x_4 + 0x_5 = 42$$

$$x_1 + x_2 + x_5 = 21$$

Identify initial basic feasible solution $x_1, x_2 = 0$

$$\text{So, } x_3 = 48, \quad \text{So, } x_3 = 48, \quad x_4 = 42, \quad x_5 = 21$$

1).

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	x_B
0	x_3	2	3	1	0	0	48
0	x_4	1	3	0	1	0	42
0	x_5	1	1	0	0	1	21
	Z_j	0	0	0	0	0	
	$Z_j - C_j$	-2	-4	0	0	0	Z_{max}

Key Column.

Calculus values of Z_j

$$0 \times 2 + 0 \times 1 + 0 \times 1 = 0$$

$$0 \times 3 + 0 \times 3 + 0 \times 1 = 0$$

$$0 \times 1 + 0 \times 0 + 0 \times 0 = 0$$

$$0 \times 0 + 0 \times 1 + 0 \times 0 = 0$$

$$0 \times 0 + 0 \times 0 + 0 \times 1 = 0$$

→ Pivot element

Next make key make zero
key column

key column $\Rightarrow \min \{-2, -4\} = -4$.

key row $\Rightarrow \min \left(\frac{18}{3}, \frac{42}{3}, \frac{21}{1} \right) = 14$

Therefore x_2 entering the basis.

$R_1' = R_1 - R_2$ $R_2' = R_2/3$ $R_3' = R_3 - 1/3 R_2$

2).

C_j		2	4	0	0	0	
C_B	B_V	x_1	x_2	x_3	x_4	x_5	x_B
0	x_3	1	0	1	-1	0	6
4	x_2	1/3	1	0	1/3	0	14
0	x_5	2/3	0	0	-1/3	1	7
	Z_j	4/3	4	0	4/3	0	
	$Z_j - C_j$	-2/3	0	0	4/3	0	2=0

calculus values of new z

$0x_1 + 4x_2/3 + 0x_3 = 4/3$

$0x_1 + 4x_2 + 0x_3 = 4$

$0x_1 + 4x_2 + 0x_3 = 0$

$0x_1 + 4x_2/3 + 0x_3 = 4/3$

$0x_1 + 4x_2 + 0x_3 = 0$

key column.

key column $\Rightarrow \min \{-2/3, 4/3\} = -2/3$.

key row $\Rightarrow \min \left(\frac{6}{1}, \frac{14}{1/3}, \frac{7}{2/3} \right) = 6$.

Therefore x_1 entering the basis. and x_3 is leaving variable.

$R_1'' = R_1'$ $R_2'' = R_2' - \frac{R_1'}{3}$ $R_3'' = R_3' - \frac{2R_1'}{3}$

3).

C_j		2	4	0	0	0	
C_B	B_V	x_1	x_2	x_3	x_4	x_5	x_B
2	$x_1 = R_1$	1	0	1	-1	0	6
4	$x_2 = R_2$	0	1	-1/3	2/3	0	12
0	$x_5 = R_3$	0	0	-2/3	1/3	1	3
	Z_j	2	4	2/3	2/3	0	
	$Z_j - C_j$	0	0	2/3	2/3	0	

$Z_j - C_j \geq 0$.

Therefore optimal solution reached

$x_1 = 6$ $x_2 = 12$ $x_5 = 3$ $x_3 = 0$ $x_4 = 0$
and $\max Z = 60$

(-) value and infinite value does not consider for key row.

~~Solve~~ Use the simplex method to solve the (LP) model.

1). $\max z = 5x_1 + 4x_2$

$$\begin{aligned} 6x_1 + 4x_2 &\leq 24 \\ x_1 + 2x_2 &\leq 6 \\ -x_1 + x_2 &\leq 1 \\ x_2 &\leq 2 \end{aligned} \quad \text{and } x_1, x_2 \geq 0$$

2). $\max z = 4x_1 - x_2$

Subjecto, $\begin{aligned} x_1 + 2x_2 &\leq 4 \\ 2x_1 + 3x_2 &\leq 3 \\ \text{and } x_1, x_2 &\geq 0 \end{aligned}$

3). $\max z = 16x_1 + 17x_2 + 10x_3$

Subject to, $\begin{aligned} x_1 + 2x_2 + 4x_3 &\leq 2000 \\ 2x_1 + x_2 + x_3 &\leq 3600 \\ x_1 + 2x_2 + 2x_3 &\leq 2400 \\ x_1 &\leq 30 \\ \text{and } x_1, x_2, x_3 &\geq 0 \end{aligned}$

4). A company makes two kinds of leather belts, belt A and belt B. Belt A is a high quality belt and belt B is of lower quality. The respective profits are Rs. 4 and Rs 3 per belt. The production of each of type A requires twice as much time as a belt of type B, and if all belts were of type B, the company could make 1,000 belts per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle and only 400 of these are available per day. There ^{are} only 700 buckles a day available for belt B. What should be the daily production of each type of belt? Formulate this problem as an LP model and solve it using the S.M.

4). $x_1 = A \quad x_2 = B. \quad \max z = 4x_1 + 3x_2$

$$\begin{aligned} 2x_1 + x_2 &\leq 1000 \quad (\text{Time availability}) \\ x_1 + x_2 &\leq 800 \quad (\text{Supply of leather}) \\ x_1 &\leq 400 \quad (\text{Buckles availability}) \\ x_2 &\leq 700 \end{aligned}$$

$$x_1, x_2 \geq 0.$$

$$\max z = 4x_1 + 3x_2 + 0S_1 + 0S_2 + 0S_3 + 0S_4$$

1). $2x_1 + x_2 + S_1 + 0S_2 + 0S_3 + 0S_4 = 1000$

2). $x_1 + x_2 + 0S_1 + S_2 + 0S_3 + 0S_4 = 800$

3). $x_1 + 0x_2 + 0S_1 + 0S_2 + S_3 + 0S_4 = 400$

4). $0x_1 + x_2 + 0S_1 + 0S_2 + 0S_3 + S_4 = 700$

1)

$$x_1, x_2, s_1, s_2, s_3, s_4 = 0$$

$$s_1 = 1000 \quad s_3 = 400$$

$$s_2 = 800 \quad s_4 = 700$$

1)

Cj	4	3	0	0	0	0	
CB BV	x_1	x_2	s_1	s_2	s_3	s_4	x_B
0	s_1	2	1	1	0	0	1000
0	s_2	1	1	0	1	0	800
0	s_3	1	0	0	0	1	400
0	s_4	0	1	0	0	0	700
Z_j	0	0	0	0	0	0	
$Z_j - C_j$	-4	-3	0	0	0	0	

key column.

$$R_1' = R_1 - 2R_3 \quad R_2' = R_2 - R_3 \quad R_3' = R_3 \quad R_4' = R_4$$

Therefore x_1 entering the basis and s_3 is leaving variable.

2)

Cj	4	3	0	0	0	0	
CB BV	x_1	x_2	s_1	s_2	s_3	s_4	x_B
0	s_1	0	1	1	0	-2	200
0	s_2	0	1	0	1	-1	400
4	x_1	1	0	0	0	1	400
0	s_4	0	1	0	0	0	700
Z_j	4	0	0	0	0	0	
$Z_j - C_j$	0	-3	0	0	0	0	

key column.

$$R_1'' = R_1' \quad R_2'' = R_2' - R_1' \quad R_3'' = R_3' \quad R_4'' = R_4' - R_3'$$

Therefore x_2 entering the basis and s_1 is leaving variable.

3)

Cj	4	3	0	0	0	0	
CB BV	x_1	x_2	s_1	s_2	s_3	s_4	x_B
3	x_2	0	1	1	0	-2	200
0	s_2	0	0	-1	1	1	200
4	x_1	1	0	0	0	1	400
0	s_4	0	0	1	0	-1	500
Z_j	4	3	3	0	-2	0	
$Z_j - C_j$	0	0	3	0	-2	0	

$$R_1''' = R_1'' + 2R_2''$$

$$R_2''' = R_2''$$

$$R_3''' = R_3'' - R_2'' \quad R_4''' = R_4'' - 2R_2''$$

Therefore s_3 entering the basis and s_2 is leaving variable.

Cj	4	
CB BV	x_1	
3	x_2	
0	s_3	
4	x_1	
0	s_4	
Z_j	4	
$Z_j - C_j$	0	

Art min

Use the min Sub

SI m

m
 x_1
 x_2
 x_3

$$\frac{1800}{800} = 2.25$$

	C_j	4	3	0	0	0	0	
CB	BV	x_1	x_2	s_1	s_2	s_3	s_4	x_B
3	x_2	0	1	-1	2	0	0	600
0	s_3	0	0	-1	1	1	0	200
4	x_1	1	0	1	-1	0	0	200
0	s_4	0	0	1	-2	0	1	100
	Z_j	4	3	1	2	0	0	
	$Z_j - C_j$	0	0	1	2	0	0	

$$(Z_j - C_j) \geq 0$$

$$\begin{cases} x_2 = 600 \\ x_1 = 200 \end{cases} //$$

$$\text{Max } z = 2600 //$$

Big M:- method. (2)

Artificial variable = $-M(x_6 + x_7 + \dots)$.

$$\min z = z \quad \max z^* = -z$$

Use penalty (Big-M) method to solve the following LP problem.

$$\min z = x_1 + x_2 + x_3$$

$$\text{Subject to, } x_1 + 2x_2 \geq 3$$

$$x_2 + 7x_3 \leq 6$$

$$x_1 - 3x_2 + 5x_3 = 5$$

$$x_1, x_2, x_3 \geq 0$$

Slack variable x_5 and surplus variable x_4 .

$$\max z = -x_1 - x_2 - x_3$$

$$x_1 + 2x_2 - x_4 = 3$$

$$x_2 + 7x_3 + x_5 = 6$$

$$x_1 - 3x_2 + 5x_3 = 5$$

$$x_1, x_2, x_3 \geq 0$$

$$\max z = -x_1 - x_2 - x_3 + 0x_4 + 0x_5 - M(x_6 + x_7)$$

$$x_1 + 2x_2 - x_4 + 0x_5 + x_6 + 0x_7 = 3$$

$$x_2 + 7x_3 + x_4 + x_5 + 0x_6 + 0x_7 = 6 \quad \leftarrow \text{(less than)}$$

$$x_1 - 3x_2 + 5x_3 + 0x_4 + 0x_5 + 0x_6 + x_7 = 5 \quad \leftarrow \text{(equal)}$$

Two-phase simplex method.

1) $\min z = 6x_1 + 21x_2$

Subject to the constraints.

$$x_1 + 2x_2 \geq 3$$

$$x_1 + 4x_2 \geq 4 \quad x_1, x_2 \geq 0$$

ts. $\leq$

$$s \geq \max z^* = -6x_1 - 21x_2$$

Such that,

Subject to the constraints

$$x_1 + 2x_2 - S_1 + A_1 = 3$$

$$x_1 + 4x_2 - S_2 + A_2 = 4$$

$$x_1, x_2, S_1, S_2, A_1, A_2 \geq 0 \quad \text{Where } z^* = -z$$

$$\max z^* = -A_1 - A_2$$

Phase Phase I

	C_j	0	0	0	0	-1	-1	
C_B	B_V	x_1	x_2	S_1	S_2	A_1	A_2	x_B
$R_1 \rightarrow$	-1	A_1	1	$\frac{1}{2}$	-1	0	1	3
$R_2 \rightarrow$	-1	A_2	1	4	0	-1	0	4
$-1 + (-1)$		Z_j	-2	-6	1	1	-1	
		$Z_j - C_j$	-2	-6	1	1	0	

key column = $\min(Z_j - C_j)$

key row = $\left(\frac{x_B}{C_B}\right)$

$$\frac{3}{2} \quad \frac{4}{4}$$

Pivot column x_2 .

basis.

Therefore x_2 is entering and A_2 is leaving variable.

$$R_1' = R_1 - (R_2/2)$$

$$R_2' = R_2/4$$

	C_j	0	0	0	0	-1	-1	
C_B	B_V	x_1	x_2	S_1	S_2	A_1	A_2	x_B
R_1'	-1	A_1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	1
	0	x_2	$\frac{1}{4}$	1	$-\frac{1}{4}$	0	$\frac{1}{4}$	1
		Z_j	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$\frac{1}{2}$	
		$Z_j - C_j$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{3}{2}$	

Pivot column S_2

Therefore s_2 entering basis and A_1 is leaving variable.
 $R_1' = 2R_1$ $R_2' = R_2 + \frac{1}{2}R_1$ $-\frac{1}{4} + \frac{1}{2} \times \frac{1}{2} = 0$

	C_j	0	0	0	0	-1	-1	
CB	BV	x_1	x_2	s_1	s_2	A_1	A_2	x_B
0	s_2	1	0	-2	1	2	-1	2
0	x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$\frac{3}{2}$
	Z_j	0	0	0	0	0	0	
	$Z_j - C_j$	0	0	0	0	1	1	$z=0$

$$0 \times 1 + 0 \times \frac{1}{2}$$

Phase II. modified simplex method.

$$\max z = -6x_1 - 21x_2$$

	C_j	-6	-21	0	0	0	
CB	BV	x_1	x_2	s_1	s_2	A_1	x_B
0	s_2	1	0	-2	1	2	2
-21	x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{3}{2}$
	Z_j	$-\frac{21}{2}$	$-\frac{21}{1}$	$\frac{21}{2}$	0	$-\frac{21}{2}$	
	$Z_j - C_j$	$-\frac{9}{2}$	0	$\frac{21}{2}$	0	$-\frac{21}{2}$	

← key row

↑
key column

$$R_1' = R_1 \quad R_2' = R_2 - \frac{R_1}{2}$$

∴ Therefore x_1 entering basis and s_2 is leaving variable

	C_j	-6	-21	0	0	0	
CB	BV	x_1	x_2	s_1	s_2	A_1	x_B
-6	x_1	1	0	-2	1	2	2
-21	x_2	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$
	Z_j	-6	-21	$\frac{3}{2}$	$\frac{9}{2}$	$-\frac{3}{2}$	
	$Z_j - C_j$	0	0	$\frac{3}{2}$	$\frac{9}{2}$	$-\frac{3}{2}$	

$(Z_j - C_j) \geq 0$ the optimal solution.

$$x_1 = 2, \quad x_2 = \frac{1}{2}, \quad s_1 = 0, \quad s_2 = 0, \quad A_1 = 0$$

2). ~~max $z = 3x_1 - x_2$~~
~~Subject to the constraints,~~
 ~~$2x_1 + x_2 - S_1 + A_1 = 2$~~
 ~~$x_1 + 3x_2 + S_2 = 2$~~
 ~~$x_2 + S_3 = 4$~~
 ~~$x_1, x_2, S_1, S_2, S_3, A_1 \geq 0$~~
~~Sur~~

max $z = 3x_1 - x_2$

Subject to,

$2x_1 + x_2 \geq 2$ ✓

$x_1 + 3x_2 \leq 2$

$x_2 \leq 4$

and $x_1, x_2 \geq 0$.

standard LPP

max $z = 3x_1 - x_2$

Subject to the constraints,

$2x_1 + x_2 - S_1 + A_1' = 2$

$x_1 + 3x_2 + S_2 = 2$

$x_2 + S_3 = 4$

max $z = -A_1$

P

C_j		0	0	0	0	0	-1	
CB	BV	x_1	x_2	S_1	S_2	S_3	A_1	x_B
-1	A_1	(2)	1	-1	0	0	1	x_B 2
0	S_2	1	3	0	1	0	0	2
0	S_3	0	1	0	0	1	0	4
	Z_j	-2	-1	1	0	0	-1	
	$Z_j - C_j$	-2	-1	1	0	0	0	

↑

key column.

1. Therefore x_1 is entering basis and A_1 is leaving variable.

$R_1' = R_1 / 2$

$R_2' = R_2 - R_1 / 2$

$R_3' = R_3$

C_j	0	0	0	0	0	0	
C_B	B.V	x_1	x_2	s_1	s_2	s_3	x_B
0	x_1	1	$1/2$	$1/2$	0	0	1/2
0	s_2	0	$1/2$	$1/2$	$1/2$	1	1/2
0	s_3	0	1	0	0	1	1
Z_j		0	0	0	0	0	
$Z_j - C_j$		0	0	0	0	0	

Phase II

C_j	3	-1	0	0	0		
C_B	B.V	x_1	x_2	s_1	s_2	s_3	x_B
3	x_1	1	$1/2$	$-1/2$	0	0	
0	s_2	0	$5/2$	$(1/2)$	1	0	2
0	s_3	0	1	0	0	1	4
	Z_j	3	$3/2$	$-3/2$	0	0	
	$Z_j - C_j$	0	$5/2$	$-3/2$	0	0	

↑
key column.

$$b'_1 = 2 - \frac{1}{2} \times 4 = 1$$

$$b'_2 = 4$$

Therefore s_1 entering basis and s_2 is leaving variable.

	C_j	3	-1	0	0	0	
C_B	B.V	x_1	x_2	s_1	s_2	s_3	x_B
3	x_1	1	3	0	1	0	2
0	s_1	0	5	1	2	0	2
0	s_3	0	1	0	0	1	4
	Z_j	3	9	0	3	0	
	$Z_j - C_j$	0	10	0	3	0	$2 + 6$

$$(Z_j - C_j) \geq 0$$

The optimal solution.

$$x_1 = 2$$

$$x_2 = -1$$

$$s_1 = 2$$

$$s_3 = 4$$

$$x_2 = 0$$

1. Transportation Problem.

North - west corner rule.

Example 1.

A	B	C	Supply.
2	7	4	5
3	3	1	8
5	4	7	7
1	6	2	14
Demand	7	9	18

$$\text{Total supply} = 5 + 8 + 7 + 14 = 34$$

$$\text{Total Demand} = 7 + 9 + 18 = 34$$

Example 2.

			Supply.
2	5	3	8
1	2	1	6
3	2	5	10
0	1	0	14
Demand	7	9	5

$$\text{Total supply} = 8 + 6 + 10 + 14 = 38$$

$$\text{Total supply} = 7 + 9 + 5 = 31$$

$$\sum_{i=1}^n a_{ij} \neq \sum_{j=1}^n b_j \quad \therefore \text{unbalance problem.}$$

Convert the balance.

2	5	3	0	8
1	2	1	0	6
3	2	5	0	10
0	1	0	0	14
7	9	15	7	

← demand column.

Example 3

1	3	4	5	10
2	1	3	0	8
5	3	2	1	7
4	2	1	3	5
5	8	12	8	

$$\sum_{i=1}^n a_i = 30$$

$$\sum_{j=1}^n b_j = 33$$

$$\sum_{i=1}^n a_i \neq \sum_{j=1}^n b_j$$

Convert the balance,

1	3	4	5	10
2	1	3	0	8
5	3	2	1	7
4	2	1	3	5
0	0	0	0	3
5	8	12	8	

← dummy row.

Example 1,

	A	B	C	Supply
1	2	7	4	5
2	3	3	1	6
5	4	4	7	4
1		6	2	14
Demand	7	9	18	14
	0	0	0	

$$\min(5, 7) = 5$$

$$\min(2, 8) = 2$$

$$\min(6, 9) = 6$$

$$\text{Total cost} = 5 \times 2 + 3 \times 2 + 3 \times 6 + 7 \times 4 + 4 \times 3 + 14 \times 2$$

Source	Destination			Supply
	A	B	C	
1	2	2	3	7
2	4	1	2	7
3	1	3	1	5
demand	20	15	30	

$$\text{Total cost} = 10 \times 2 + 10 \times 4 + 10 \times 3 + 30 \times 1 + 5 \times 1 = 125$$

Source	Destination			Supply
	A	B	C	
1	⁽¹⁵⁾ 2	7	4	50 50
2	⁽³⁾ 3	⁽⁶⁾ 3	1	8 80
3	5	⁽⁴⁾ 4	⁽¹⁾ 7	70
4	1	6	⁽¹²⁾ 2	140
Demand	7	9	18	
	2 0	3 0	1 0	

$$\text{Total cost} = 5 \times 2 + 3 \times 2 + 6 \times 3 + 3 \times 4 + 7 \times 7$$

$$= 102$$

least cost Method.

Source	Destination			Supply
	A	B	C	
1	2	⁽⁴⁾ 7	4	8
2	3	3	⁽¹⁾ 1	8 20
3	5	⁽⁴⁾ 4	7	8 0
4	⁽¹⁾ 1	6	⁽²⁾ 2	2 70
Demand	7	9	18	140
	0	0	0	

$$\text{Total cost} = 7 \times 2 + 4 \times 3 + 8 \times 1 + 4 \times 7 + 50$$

$$= 7 \times 2$$

$$= 14$$

very good

10
36
18
12
28
28
102

40
90
50
190
180
450

50
40
90
180
180
40

14
12
28
28
14

from 1 to 2

Solve the transportation method problem.

Source

	Destination				Supply
	A	B	C	D	
1	5	4	2	6	20
2	8	3	5	7	30
3	10	9	1	6	50
Demand	70	40	20	30	100

Least cost method.

$$\begin{aligned} \text{Total cost} &= 20 \times 2 + 30 \times 3 + 5 \times 10 + 9 \times 10 + 6 \times 30 \\ &= \underline{\underline{450}} \end{aligned}$$

$$\begin{array}{r} 20 \\ 60 \\ 50 \\ 90 \\ 30 \\ \hline 450 \end{array}$$

100	5	4	2	6	20	70	0
100	8	3	5	7	30	0	
100	5	9	4	6	50	30	0
100	10	40	20	30			

North east west corner.

$$\begin{aligned} \text{Total cost} &= 10 \times 5 + 10 \times 4 + 30 \times 3 + 20 \times 4 + 30 \times 6 \\ &= \underline{\underline{440}} \end{aligned}$$

$$\begin{array}{r} 50 \\ 40 \\ 90 \\ 20 \\ 180 \\ \hline 400 \end{array}$$

Find	solution using North - west corner method.			
	D ₁	D ₂	D ₃	D ₄
S ₁	9	30	50	10
S ₂	70	30	40	60
S ₃	40	8	70	20
Demand	5	8	7	14

$$\text{Total Supply} = \sum_{i=1}^3 a_i = 7 + 9 + 18 = 34$$

$$\text{Total Demand} = \sum_{i=1}^4 b_i = 5 + 8 + 7 + 14 = 34$$

$$\text{Total supply} = \text{Total demand} = 34$$

1). First Allocation. (transportation).

	D ₁	D ₂	D ₃	D ₄	Supply
S ₁	19 5	30	50	10	78 20
S ₂	70	30 ⁶	40 ³	60	78 30
S ₃	40	8	70 ⁴	20 14	18 160
Demand	50	8 20	7 40	140	

transportation.

$$\begin{aligned} \text{Total cost} &= 19 \times 5 + 30 \times 2 + 30 \times 6 + 40 \times 3 + \\ &\quad 70 \times 4 + 20 \times 14 \\ &= \underline{1015} \end{aligned}$$

2). Find the solution least cost method.

	D ₁	D ₂	D ₃	D ₄	Supply.
S ₁	19	30	50	10 ⁷	70
S ₂	70 ²	30	40 ⁷	60	920
S ₃	40 ³	8 ⁸	70	20 ⁷	18 1030
Demand.	50	80	70	1470	

$$\begin{aligned} \text{Total least cost} &= 70 \times 2 + 40 \times 3 + 8 \times 8 + \\ &\quad 40 \times 7 + 10 \times 7 + 20 \times 7 \\ &= \underline{814} \end{aligned}$$

1). North-west corner method.

	D ₁	D ₂	D ₃	D ₄	Supply.
S ₁	18	8	4	3	500
S ₂	12	14	20	2	400
S ₃	6	9	28	25	300
Demand	250	350	600	150	

$$\text{Total Supply } \sum_{i=1}^3 a_i = 500 + 400 + 300 = 1200$$

$$\text{Total demand } \sum_{i=1}^4 b_i = 250 + 350 + 600 + 150 = 1350$$

Total Supply $\neq$ Total demand.

This problem is unbalance.

	D ₁	D ₂	D ₃	D ₄	Supply
S ₁	18	8	4	3	500
S ₂	12	14	20	2	400
S ₃	6	9	23	25	300
Demand	250	350	600	150	

1).

	D ₁	D ₂	D ₃	D ₄	Supply
S ₁	18 ²⁵⁰	8 ²⁵⁰	4	3	500 - 250 = 0
S ₂	12	14 ¹⁰⁰	20 ³⁰⁰	2	400 - 300 = 0
S ₃	6	9	23 ³⁰⁰	25	300 - 0 = 300
Demand	250	350	600	150	150 - 0 = 150

Total transportation cost =

$$= 18 \times 250 + 8 \times 250 + 14 \times 100 + 20 \times 300 + 23 \times 300$$

$$= \underline{\underline{20,800}}$$

2).

	D ₁	D ₂	D ₃	D ₄	Supply
S ₁	18	8	14 ⁵⁰⁰	3	500 - 0 = 500
S ₂	12	14 ²⁵⁰	20	2 ¹⁵⁰	400 - 150 = 0
S ₃	6 ²⁵⁰	9 ⁵⁰	23	25	300 - 50 = 250
Demand	250	350	600	150	150 - 150 = 0

Total least cost = $4 \times 500 + 14 \times 250 + 2 \times 150 + 6 \times 250 + 9 \times 50$

$$= \underline{\underline{7,750}}$$

Vogel's Approximation Method (VAM).

	D ₁	D ₂	D ₃	D ₄	Supply	P ₁	row penalty
S ₁	19	30	50	10	7	9	
S ₂	70	30	40	60	9	10	
S ₃	40	8	70	20	18	12	
Demand	5	8	7	14			
P ₁	21	22	10	10			

column penalty.

	D ₁	D ₂	D ₃	D ₄	Supply	P ₁	P ₂
S ₁	19	30	50	10	7	9	9
S ₂	70	30	40	60	9	10	20
S ₃	40	8 ^(cs)	70	20	18	12	20
Demand	5	8	7	14			
P ₁	21	22	10	10			
P ₂	21	-	10	10			

Maximum penalty value. minimum unit cost.

	D ₁	D ₂	D ₃	D ₄	Supply	P ₁	P ₂	P ₃	P ₄
S ₁	19 ⁵	30	50	10 ²	7	20	9	40	40
S ₂	70	30	40 ¹	60 ²	9	10	20	20	20
S ₃	40	8 ⁸	70	20 ¹⁰	18	12	20	50	-
Demand	5	8	7	14					
P ₁	21	22	10	10					
P ₂	21	-	10	10					
P ₃	-	-	10	10					
P ₄	-	-	-	-					

$$\begin{aligned}
 \text{Total cost} &= 19 \times 5 + 10 \times 2 + 40 \times 7 + 60 \times 2 + 8 \times 8 + 20 \times 10 \\
 &= 779
 \end{aligned}$$

Example 2

Unbalance Problem.

	D ₁	D ₂	D ₃	D ₄	Supply	P ₁	P ₂	P ₃	P ₄
S ₁	18	8	4 ⁵⁰⁰	3	500	1	4	4	-
S ₂	12	14	20	2 ¹⁵⁰	400	10	2	2	2
S ₃	6	9	23	25	300	3	3	3	3
	0	0 ¹⁵⁰	0	0	150	0	-	-	-
Demand	250	350	600	150					
P ₁	6	8	4	2					
P ₂	6	8	4	-					
P ₃	6	1	16	-					
P ₄	6	5	3	-					

	D ₁	D ₂	D ₃	D ₄	Supply	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆
S ₁	18	8	4 ⁵⁰⁰	3	500	1	4	4	-	-	-
S ₂	12	14 ¹⁵⁰	20 ¹⁰⁰	2 ¹⁵⁰	400	10	2	2	2	3	3
S ₃	6 ²⁵⁰	9 ⁵⁰	23	25	300	3	3	3	3	2	-
	0	0 ¹⁵⁰	0	0	150	0	-	-	-	-	-
Demand	250	350 ²⁵⁰	600 ¹⁰⁰	150							
P ₁	6	8	4	2							
P ₂	6	8	4	-							
P ₃	6	1	16	-							
P ₄	6	5	3	-							
P ₅	-	5	3	-							
P ₆	-	14	20	-							

$$\text{Total cost} = 4 \times 500 + 14 \times 150 + 20 \times 100 + 2 \times 150 + 6 \times 250 + 9 \times 50 + 30 \times 30 = 3500$$

Modified Distribution Method. (MODI)

Example:-

Consider the transportation problem given below

Apply the least cost method. (Modimethod)

	U_1	U_2	U_3	Supply
U_1	8^{*30}	9	7^{*10}	40
U_2	4	3^{*25}	5	25
U_3	8	5^{*5}	6^{*30}	35
Demand	30	30	40	

2

$$x_{11} = 30, \quad x_{13} = 10, \quad x_{22} = 25 \\ x_{23} = 5, \quad x_{33} = 30$$

The dual variable U_1, U_2, U_3 and V_1, V_2, V_3 can be calculated from the corresponding C_{ij} values, that is:
 $U_1 + V_1 = 8, \quad U_1 + V_3 = 7, \quad U_2 + V_2 = 3, \quad U_2 + V_3 = 5, \quad U_3 + V_3 = 6$
 $U_1 + V_2 = 6, \quad U_3 + V_1 = 8$
 $U_1 = 0, \quad U_2 = -3, \quad U_3 = -1$
 $V_1 = 8, \quad V_2 = 6, \quad V_3 = 7$

Choose any one of the dual variables arbitrary zero. Let $U_1 = 0$

$$V_1 = 8, \quad V_2 = 6, \quad U_2 = -3, \quad V_3 = 7, \quad U_3 = -1$$

	$U_1 = 0$	$U_2 = -3$	$U_3 = -1$	Supply
$U_1 = 0$	8^{*30}	9	7^{*10}	40
$U_2 = -3$	4	3^{*25}	5	25
$U_3 = -1$	8	5^{*5}	6^{*30}	35
Demand	30	30	40	

Now we calculate $C_{ij} - U_i - V_j$ values for all the cells where $x_{ij} = 0$ (i.e. unallocated cell by the basic feasible solution). That is,

$$\text{cell } (1,2) = C_{12} - U_1 - V_2 = 9 - 0 - 6 = 3$$

$$\text{cell } (2,1) = C_{21} - U_2 - V_1 = 4 - (-3) - 8 = -1$$

$$\text{cell } (2,3) = C_{23} - U_2 - V_3 = 5 + 3 - 7 = 1$$

$$\text{cell } (3,1) = C_{31} - U_3 - V_1 = 8 + 1 - 8 = 1$$

(Note: For cell (1,2) $C_{12} = 9$, $U_1 = 0$, $V_2 = 6$)

(10-11-2020)

Note that in the above calculation all the $c_{ij} - u_i - v_j \geq 0$ except for cell (2,1) where,

$$C_{21} - u_2 - v_1 = 4 + 3 - 8 = -1$$

	$v_1 = 8$	$v_2 = 6$	$v_3 = 7$	Supply.
$u_1 = 0$	8 * 30	9	7 * 10	40
$u_2 = -3$	4 * 25	3 * 25	5	25
$u_3 = -1$	8	5 * 5	6 * 30	35
Demand.	30	30	40	

Select the minimum value from allocated values to the cell with a minus sign. Subtract this value from the cell with a minus sign and add to the cell with a plus sign.

	$v_1 = 8$	$v_2 = 6$	$v_3 = 7$	Supply.
$u_1 = 0$	8 * 5	9	7 * 35	40
$u_2 = -3$	4 * 25	3	5	25
$u_3 = -1$	8	5 * 30	6 * 5	35
Demand.	30	30	40	

$$\text{Optimal cost} = 8 \times 5 + 7 \times 35 + 4 \times 25 + 5 \times 30 + 6 \times 5$$

$$= \underline{\underline{565}}$$

Hungarian Method Example.

		Machine.			
J		1	2	3	
O	1	5	7	9	1
B	2	14	10	12	1
	3	15	13	16	1
		1	1	1	

Step 1: Select the smallest value in each row.

J		1	2	3	
O	1	[5]	7	9	1
B	2	14	[10]	12	1
	3	15	[13]	16	1
		1	1	1	

J		1	2	3	
O	1	0	2	4	1
B	2	4	0	[2]	1
	3	2	0	3	1

~~Step 2:~~

J		1	2	3	
O	1	0	2	2	1
B	2	4	0	0	1
	3	2	0	1	1

Step 2: Do the same for the columns that do not have any zero value.

		Machine.			
J		1	2	3	
O	1	0	2	<u>2</u>	1
B	2	4	0	0	1
	3	2	0	1	1

step 3: Assignments are made at zero values.
Therefore, we assign job 1 to machine 1; job 2 to machine 3, and job 3 to machine 2.

Total cost = $5 + 13 + 12 = 30$

It is not always possible to obtain a feasible assignment as in here.

Example 2

row
Subordinates.

T		1	2	3	4
A	A	(8)	26	17	11
S	B	13	28	(4)	26
K	C	38	19	18	(15)
S	D	19	26	24	(10)

row

Subordinates.

T		1	2	3	4
A	A	0	18	9	3
S	B	9	24	0	22
K	C	23	14	3	0
S	D	9	26	14	0

column

Subordinates.

T		1	2	3	4
A	A	0	16	9	3
S	B	9	20	0	22
K	C	23	0	3	0
S	D	9	12	14	0

Optimal assignment.

$A \rightarrow 1, B \rightarrow 3, C \rightarrow 2, D \rightarrow 4$

Total man-hours = $8 + 4 + 19 + 10 = \underline{\underline{41h}}$

Unbalanced assignment problem.
Example:

Solve the following unbalanced assignment problem of minimizing the total time for performing all the jobs.

Jobs.

W		1	2	3	4	5	6
0	A	5	2	4	2	5	0
1	B	2	4	7	6	6	0
k	C	6	7	5	8	7	0
E	D	5	2	3	3	4	0
r	E	8	3	7	8	6	0
S.	F	3	6	3	5	7	0

Step 1: Select the smallest value in each row and subtract this value from each value in that row.

	1	2	3	4	5	6
A	5	<u>2</u>	4	<u>2</u>	5	0
B	<u>2</u>	4	7	6	6	0
C	6	7	5	8	7	0
D	5	<u>2</u>	<u>3</u>	3	<u>4</u>	0
E	8	3	7	8	6	0
F	3	6	<u>3</u>	5	7	0

Does not change.

Step 2: Select the smallest value in each column and subtract this value from each value in that column.

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	1	2	3	4	5	6
A	3	2	1	0	1	0
B	0	2	4	4	2	0
C	4	5	2	6	3	0
D	3	2	1	1	0	0
E	6	1	4	6	2	0
F	1	4	0	3	3	0

	1	2	3	4	5	6
A	3	0	1	0	1	0
B	0	2	4	4	2	0
C	4	5	2	6	3	0
D	3	0	0	1	0	0
E	6	1	4	6	2	0
F	1	4	0	3	3	0

row
ch

	1	2	3	4	5	6
A	3	0	1	0	1	0
B	0	2	4	4	2	0
C	4	5	2	6	3	0
D	3	0	0	1	0	0
E	5	0	3	5	1	0
F	1	4	0	3	3	0

$$A4 + B1 + D5 + E2 + F3$$

The total minimum time is = $2 + 2 + 4 + 3 + 3$
= 14.

Example,

Jobs
e in

A company has 4 machines on which to do 3 jobs. Each job can be assigned to one and only one machine. The cost of each job on each machine is given in the following table.

	w	x	y	z
A	18	24	28	32
B	8	13	14	19
C	10	15	19	22

Machines .

		W	X	Y	Z
Jobs .	A	18	24	28	32
	B	8	13	17	19
	C	10	15	19	22
	D	0	0	0	0

Now, the problem is balanced.

Row		W	X	Y	Z
	A	0	6	10	14
	B	0	5	9	11
	C	0	5	9	12
	D	0	0	0	0

2nd Modified matrix.

0	6-5	10-5	14-5	0	1	5	9
0	5-5	9-5	11-5	0	0	4	6
0	5-5	9-5	12-5	0	0	4	7
0	0	0	0	0	0	0	0

3rd Modified matrix.

0	1	5-4	9-4	0	1	1	5
0	0	0	6-4	0	0	0	2
0	0	0	7-4	0	0	0	3
0	0	0	0	0	0	0	0

Now, optimal solution

$$A \rightarrow W + B \rightarrow X + C \rightarrow Y + Z \rightarrow D$$

$$= 18 + 13 + 19 = 50$$

~~Ha~~ Mettalica works Ltd. Is a metal working machine shop. It generally does work on contract basis. At present it has for jobs A, B, C, D in hand Mettalica for machines - 1, 2, 3 and 4. The jobs can be performing a job on different machines vary ~~because~~ because of

m is
ed.

capacity machine and characteristics of the job. The jobs are to be exclusively performed on any one machine as setup cost is too expensive to permit setting of an other machine for same job. An estimate has been made of the cost of performing the job on each machine and estimated cost total is tabulated below.

Job/Machine.	1	2	3	4
A	20	36	31	17
B	24	32	40	12
C	22	40	38	18
D	36	39	35	16

ist) Row Reduced Matrix,

	1	2	3	4
A	3	19	14	0
B	12	20	28	0
C	4	22	20	0
D	20	23	19	0

iind) Column Reduced Matrix.

	1	2	3	4
A	0	0	0	0
B	9	1	14	0
C	1	3	6	0
D	17	4	5	0

3rd) Modified matrix.

	1	2	3	4
A	0	0	0	0
B	8	0	13	0
C	0	2	5	0
D	16	3	4	0

Now, Optimal Sol.ⁿ

$$A_3 + B_2 + C_1 + D_4$$

$$= 31 + 32 + 22 + 16$$

$$= \underline{\underline{101}}$$

ing
on

1, 2, 3
on